Day 23 Assignment

Description:
Built RESTful course management APIs with validation and rate limiting to improve reliability and security. This day focused on structured API design and request control.

Challenge 1 - CRUD API Setup:
Created REST API endpoints for creating, reading, updating, and deleting courses using an in-memory array. This demonstrated the core CRUD workflow in Express.

```
let courses = [
    { id: 1, name: "React Mastery", duration: "6 weeks" },
    { id: 2, name: "Node.js Basics", duration: "4 weeks" }
];

app.get("/api/v1/courses", (req, res) => {
    res.json(courses);
});

app.post("/api/v1/courses", (req, res) => {
    const newCourse = {
        id: courses.length + 1,
        name: req.body.name,
        duration: req.body.duration
    };
    courses.push(newCourse);
    res.status(201).json(newCourse);
});

app.put("/api/v1/courses/:id", (req, res) => {
    const id = parseInt(req.params.id);
    const course = courses.find(c => c.id === id);

    if (!course) {
        return res.status(404).json({ error: "Course not found" });
    }

    course.name = req.body.name || course.name;
    course.duration = req.body.duration || course.duration;

    res.json(course);
});

app.delete("/api/v1/courses/:id", (req, res) => {
    const id = parseInt(req.params.id);
    courses = courses.filter(c => c.id !== id);
    res.json({ message: "Course deleted successfully" });
});

app.listen(4000, () => {
```

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Challenge 2 - Input Validation

Validated incoming course data using express-validator before creating a new course.
This ensured required fields were checked before processing requests.

```
app.post(
    "/api/v1/courses",
    [
        body("name").notEmpty().withMessage("Course name is required"),
        body("duration").notEmpty().withMessage("Course duration is required")
    ],
    (req, res) => {
        const errors = validationResult(req);

        if (!errors.isEmpty()) {
            return res.status(400).json({ error: errors.array()[0].msg });
        }

        const newCourse = {
            id: courses.length + 1,
            name: req.body.name,
            duration: req.body.duration
        };

        courses.push(newCourse);
        res.status(201).json(newCourse);
    }
);
```

Challenge 3 - API Rate Limiting

Used express-rate-limit to restrict users to 5 requests per minute.
This helped prevent API abuse and excessive traffic.

```
const express = require("express");
const { body, validationResult } = require("express-validator");
const rateLimit = require("express-rate-limit");
const app = express();

app.use(express.json());

const limiter = rateLimit({
  windowMs: 60 * 1000,
  max: 5,
  message: { error: "Too many requests" }
});

app.use("/api/v1/", limiter);
```

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